

Mathematics 629 — Homework 8

Due Monday, April 27

Alexander Kim

Problem 1. Let $(X, \mathcal{M}, \mu)$ be a measure space, let $f : X \rightarrow [0, \infty]$ be a nonnegative $\mathcal{M}$ -measurable function and define for $\alpha > 0$

$$\lambda_f(\alpha) = \mu(\{x : f(x) > \alpha\}).$$

Show that

$$\int f(x)^p d\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) d\alpha.$$

Hint: First assume that X is σ -finite and find a way to apply Tonelli's theorem. Then extend to the general case.

Proof. First suppose X is σ -finite and let $A = \{x; f(x) > \alpha\}$ for $\alpha > 0$. Then since $f(x)^p = \int_0^{f(x)} p\alpha^{p-1} d\alpha = \int_0^\infty p\alpha^{p-1} \mathbf{1}_A d\alpha d\mu$ we can apply Tonelli's theorem to

$$\int_X f(x)^p d\mu = \int_X \int_0^\infty p\alpha^{p-1} \mathbf{1}_A d\alpha d\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) d\alpha.$$

In the general case where σ -finiteness does not hold, suppose that since $A \subset X$ then there exists $A_i = \{x; f(x) > \frac{\alpha}{i}\}$ such that $A = \cup_{i=1}^\infty A_i$. Then if any $\mu(A_j) = \infty$ it follows that $\mu(A) = \infty$ also. Thus because the integral of infinity is finite

$$\int_X f(x)^p d\mu = \infty = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) d\alpha.$$

If $m(A)$ is σ -finite then X is σ -finite and we're done. □

Problem 2. For $n + 1$ vectors $u_0, u_1, \dots, u_n \in \mathbb{R}^n$ define the simplex with vertices u_i by

$$S = \left\{ \sum_{i=0}^n \tau_i u_i : \tau_i \in [0, 1], \sum_{i=0}^n \tau_i = 1 \right\}.$$

Let B be the matrix with column vectors $u_1 - u_0, \dots, u_n - u_0$. Prove that

$$m(S) = \frac{|\det B|}{n!}.$$

Solution. Let $\tau_0 = 1 - \sum_{i=1}^n \tau_i$. Then we know for $x \in S$ that $x = \mu_\tau + \sum_{i=1}^n \tau_i (u_i - u_0) = u_0 - B\tau$ where $\tau = (\tau_1, \dots, \tau_n)^T$. Now let $l_n = \{\tau \in \mathbb{R}^n; \tau_i \in [0, 1]; \sum_{i=1}^n \tau_i \leq 1\}$ and let $B(l_n) = B\tau$ be a linear transformation in $\mathbb{R}^n$ such that $S = u_0 + B(l_n)$. By the change of variables theorem,

$$m(s) = m(B(l_n)) = |\det B| m(l_n).$$

Now we will use induction to show $m(l_n) = \frac{1}{n!}$. Let $l_1 = \{\tau \in \mathbb{R}; \tau \in [0, 1]\}$ be the base case. Clearly $m(l_n) = \frac{1}{1!}$. Suppose the inductive hypothesis holds for $n - 1$ dimensions. Then for $\tau_n = b \in [0, 1]$ it

follows that $\{(\tau_1, \dots, \tau_{n-1})^T; \tau_i \geq 0; \sum_{i=1}^{n-1} \tau_i \leq 1-b\} = (1-b)l_{n-1}$. Since $m((1-b)l_{n-1}) = \frac{(1-b)^{n-1}}{(1-n)!}$ then by the construction of the product measure of $\mathbb{R}^n$,

$$m(l_n) = \int_0^1 \frac{(1-b)^{n-1}}{(n-1)!} db = \frac{1}{(n-1)!} \frac{1}{n} = \frac{1}{n!}.$$

Thus we have shown $m(S) = \frac{|det B|}{n!}$.

Problem 3. Let $f : [0, \infty) \rightarrow \mathbb{C}$ be continuous and $\alpha > 0$. Define the α th fractional integral by

$$I^\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-s)^{\alpha-1} f(s) ds, \quad x > 0.$$

Prove that $I^{\alpha+\beta} f = I^\alpha(I^\beta f)$ for any $\alpha > 0, \beta > 0$.

Show that for $n \in \mathbb{N}$ the n th fractional integral satisfies

$$(I^n f)^{(n)}(x) = f(x),$$

i.e. $I^n f$ is an antiderivative of order n .

Remark: Study and use some facts about the Γ -function in the notes.

Proof. First observe by definition of the Γ function that

$$I^\alpha(I^\beta f(x)) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x f(t) \int_t^x (x-s)^{\alpha-1} (s-t)^{\beta-1} ds dt.$$

Let $s = t + (x-t)u$ for $u \in [0, 1]$ and $ds = (x-t)du$. Then by u-substitution,

$$I^{\alpha+\beta} f(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x (x-t)^{\alpha+\beta-1} f(t) \int_0^1 u^{\beta-1} (1-u)^{\alpha-1} du dt.$$

Then by theorem 3.21 of the Beta function,

$$\frac{1}{\Gamma(\alpha)\Gamma(\beta)} B(\alpha, \beta) \int_0^x (x-t)^{\alpha+\beta-1} f(t) \int_0^1 u^{\beta-1} (1-u)^{\alpha-1} du dt = I^{\alpha+\beta} f(x)$$

and we are done.

To show the $I^n f$ is the anti derivative of order n , let $n = 1$ be the base case and observe by definition of the Γ function that $I^1 f = \int_0^x f(s) ds = f(x)$ so the base case holds. Since f is continuous suppose this holds for $n-1$ fractional steps such that $I^n f = I^1(I^{n-1} f)$ and $(I^n f)' = I^{n-1} f$. By the inductive hypothesis then $(I^n f)^n = ((I^n f)')^{n-1} = (I^{n-1} f)^{n-1} = f$. $\square$

Problem 4. Let

$$G = \left\{ (x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, 0 < x_2 < \frac{x_1^2}{4} \right\}$$

and

$$\Omega = \{(s, t) \in \mathbb{R}^2 : 0 < s < t\}.$$

Prove that for nonnegative Lebesgue measurable f and for $f \in L^1(\mathbb{R})$ that

$$\int_G f(x_1, x_2) dm(x_1, x_2) = \int_\Omega f(s+t, st) |s-t| dm(s, t)$$

and

$$\int_G f(x_1, x_2) dm(x_1, x_2) = \frac{1}{2} \int_{(0, \infty) \times (0, \infty)} f(s+t, st) |s-t| dm(s, t).$$

Proof. First let $\phi(s, t) : \Omega \rightarrow \mathbb{R}^2 = (s + t, st)$ be a bijective linear map where $\phi(\Omega) \subset G$. Then compute $|\det \phi'| = |\det \begin{bmatrix} 1 & 1 \\ s & t \end{bmatrix}| = |s - t|$. By the linear change of variables theorem,

$$\int_G f(x_1, x_2) dm = \int_\Omega f(\phi(s, t)) |\det \phi'| dm = \int_\Omega f(s + t, st) |s - t| dm.$$

Let Ω' be the reflection of Ω on $D = \{t = s > 0\}$. Then $Q = (0, \infty) \times (0, \infty) = \Omega' D \cup \Omega$. Note that $m(D) = 0$ in $\mathbb{R}^2$. Notice that the bijective mapping $\sigma(s, t) = (t, s)$ is also measure preserving since it is an elementary matrix operation, thus

$$\int_{\Omega'} f(t + s, ts) |t - s| dm = \int_\Omega f(s + t, st) |s - t| dm$$

so by substitution

$$\int_Q f(s + t, st) |s - t| dm = 2 \int_\Omega f(s + t, st) |s - t| dm.$$

By the previously proven proposition above,

$$\int_G f(x_1, x_2) dm = \frac{1}{2} \int_Q f(s + t, st) |s - t| dm.$$

□

Problem 5. Let $E = \{x \in \mathbb{R}^n : |x| \leq 10\}$ and

$$J_b = \int_E |x_1^2 + \cos^2(x_2^4 + \cdots + x_n^4) - 2|^{-b} dm(x).$$

For which $b > 0$ is J_b finite?

Solution J_b is finite for $0 < b < 1$. If we use Tonelli's theorem, the integrand is positive for $b \geq 1$ which forces $J_b = \infty$. (ran out of type to finish type setting)